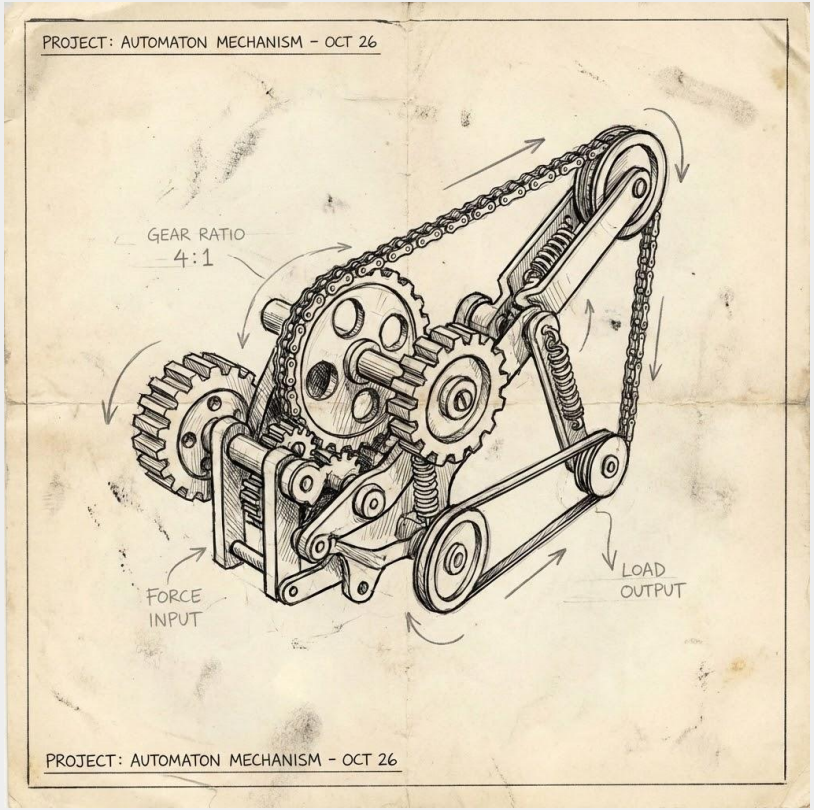
CSC3041F

Control

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Image: Lucie de Moyencourt



Control Systems

Control Systems

- What are control systems?
  - A Control System is a framework - which may be mathematical, mechanical, or electronic - used to manage, command, direct, or regulate the behaviour of other devices or systems.
- What are the components of control systems?
  - The Plant (or Process):
    - The physical system you are trying to influence (e.g., a drone's airframe, a robotic etc.).
  - The Sensor:
    - The "eyes" that measure the current state (e.g., GPS, accelerometers, or thermometers).
  - The Controller:
    - The "brain" or algorithm that decides what to do based on the sensor data, in a feedback control system.
  - The Actuator:
    - The "muscles" that carry out the controller's command (e.g., motors, valves, or heaters).

Classifications: Open and Closed Loop Control Systems

- **Open-loop Control Systems**, also known as a non-feedback controller.
  - **Logic:** Input → Controller →Actuator →Plant → Output
  - No circular path.
  - Disadvantage and motivation for closed-loop control systems:
    - Cannot handle external factors/noise/changes in the system

- **Closed-loop Control Systems**, also known as feedback control systems.
  - Unlike an open-loop system, it constantly monitors the "actual" state of the system and compares it to the "desired" state (the set-point). The system then calculates the difference between these two - the error - and takes action to reduce that error to zero.
  - The process is circular. Information "feeds back" from the output to the input stage to ensure accuracy and stability.

Open and Closed Loop Control Systems





Open loop control systems

No feedback, fixed operation based on input





Closed loop control systems

Uses feedback to adjust output and maintain desired state

Classifications: Model-based and Model-free Control

Model-based Control:

- A model-based controller relies on an explicit mathematical description of the system's (plant's) dynamics - the Model.
- Differential equations describe how the state changes over time.
- Example: Model-Predictive Control (MPC)
- Can be seen as the "Planner" control method.

Model-free Control:

- A model-free controller does not know (or care) about the underlying physics of the system. It simply maps a measured error or state directly to a control action based on a fixed rule or a learned policy.
-
- Model-Free is great for complex tasks where the physics are too hard to write down and you have the computational power to simulate millions of trials.

Model-Based Control

Advantages

- **Efficiency:** Since the physics are already known, the system doesn't need millions of "trials" to understand the likes of gravity or friction.
- **Predictability:** It can "look ahead" and anticipate corners or obstacles before they are reached.
- **Constraint Handling for Safety:** You can hard-code physical limits (e.g., "never exceed 80° C") directly into the model's logic.
- **Transparent and Interpretable:** If something goes wrong, you can check the math to see exactly why the controller made a specific decision.
- **Physics-based hence can be transferable:** A model of a motor works for any motor of that same specification without "re-training."

Disadvantages

- If the model is wrong (Model Mismatch), the controller can perform poorly or even become unstable
- **Computationally Heavy**: Solving complex optimisation problems in real-time requires significant processing power.
- **Complexity**: Writing down the exact differential equations for a complex systems, multi-jointed robot is extremely difficult.
- They often struggle with "un-modelled dynamics" - things like a sudden change in wind etc (disturbances)
- **Maintenance:** The model must be constantly updated as parts wear down or environmental conditions change.

Model-Free Control

Advantages

- **Versatility:** General Purpose. Can be applied to almost any system (games, robotics, trading) without needing to know the underlying physics.
- **Development Time:** Fast Prototyping. You don't need a team of physicists to derive differential equations.
- **Constraint Handling for Safety:** You can hard-code physical limits (e.g., "never exceed 80°C") directly into the model's logic.
- **Safety:** Autonomous Discovery. Can find creative solutions to problems that a human designer might never have thought of.

Disadvantages

- **Versatility:** Dependent on exposure in learning stage.
- It requires a massive number of interactions (millions of frames/steps) to learn even simple behaviours, if using a learning algorithm.
- **Complexity**: Writing down the exact differential equations for a complex systems, multi-jointed robot is extremely difficult.
- **Safety:** Unsafe Exploration. During the learning phase, the agent must fail to learn. In the real world, this means crashing cars or breaking robot arms.

PID Control

- PID: Proportional-Integral-Derivative

- Model-free

- Closed-loop feedback mechanism
  - that calculates an "error" value as the difference between a desired set-point and a measured process variable.
  - Decisions based on the Present, the Past, and the Future.

PID Control: Example

- Consider a cruise control system in a vehicle

- Consider a Water Tank Level Controller

PID Controller Characteristics

- SISO vs MIMO
  - Single-input single-output vs Multi-Input Multi-Output
  - PID: designed to track *one* variable
- Linear vs. Non-Linear Dynamics
  - Assumes the world is mostly linear. It works best near a "set-point." If the system moves far away from that point (e.g., a drone flipping upside down), the PID gains often become useless.
- Tracking vs. Optimisation
  - It's only goal is to minimise error.
  - It doesn't care how it gets there, as long as the error goes to zero. It has no concept of "efficiency".
- Adaptation to Change
  - Once tuned, the gains are fixed. If the robot's arm gets heavier (e.g., it picks up a box) or a motor wears out, the PID controller must be manually re-tuned.

PID Controller Drawbacks

- Still highly relies on human intuition/physics
- Decision speed*
- If complex or uncertainty, may not be the best controller.

What’s next?

What’s next?

